

Noah Takahashi

Embedded Systems & IoT Firmware Engineer • 4 Years Experience • MERIQ Index: 68% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: FreeRTOS, I2C/SPI/UART, PCB Debugging, Embedded Linux, Python, ARM Cortex, C/C++, MQTT

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core FreeRTOS service, reducing latency by 51%.
- Led a squad of 8 engineers delivering scalable microservices processing 39M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 48%.

Associate Embedded Systems & IoT Firmware Engineer — *Intel* (2022 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using PCB Debugging saving \$10k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [FreeRTOS, I2C/SPI/UART, PCB Debugging, Embedded Linux]

Engineered end-to-end production solution leveraging FreeRTOS and I2C/SPI/UART with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [PCB Debugging, Embedded Linux, Python, ARM Cortex]

Architected high-resilience system integrating PCB Debugging with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

Vellore Institute of Technology (VIT) — 2018 - 2022 (Score: 8.49 / 10.0)